25900510 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMAT204 - ESSENTIALS OF FINANCIAL MATHEMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

{Short Answer Questions}

Answer any **5** questions

Each question carries **2** marks

1. Define Geometric series. [K] / [CO1]

2. Define the term Discount rate . [U] / [CO1]

3. Find the break- even point for the following firms [U] / [CO2]
   $R(x) = 37.5x$
   $C(x) = 22.25x + 1586$

4. Define overconstrained system of equations. [K] / [CO2]

5. Convert the inequality $9x + 2y \leq 86$ into equation by introducing an appropriate slack or surplus variable. State whether the variable introduced is a slack variable or a surplus variable. [U] / [CO3]

6. Write the standard form or general form of LPP. [U] / [CO3]

7. Find critical value of marginal cost function for the total cost $TC = Q^3 - 3Q^2 + 2Q - 4$. [A] / [CO4]

8. Find the critical points of the function $f(x) = 2x^3 + 9x^2 + 12x + 10$. [A] / [CO4]

[2x5 = 10]

Part B

{Short Essay Questions}

Answer any **5** questions

Each question carries **6** marks

9. A person saves $100 in a bank account at the beginning of each month. The bank offers a return of 12% compounded annually.After how many months does the amount saved first exceed $3000 ? [E] / [CO1]

10. Find the value, in 10 years' time, of $1000 invested at 8% interest compounded annually. [A] / [CO1]

11. Find a) the reduced form b) the value of the equilibrium level of income for the following model [U] / [CO2]

$Y = C + I \qquad C = C_0 + bY$

$I = I_0 + aY \qquad and \qquad C_0 = 75 \qquad I_0 = 75$

$, b = 0.6 \qquad and \qquad a = 0.2$

12. Solve the system of equations using graphs. [U] / [CO2]

$4x - 3y = 22$

$7x - 6y = 34$

13. Find the equilibrium price and quantity for the following problems by using graphs [U] / [CO2]

$Supply : P = \frac{3}{8}Q + 1$

$Demand : P = -\frac{1}{2}Q + 22$

14. Identify the feasible region of the LPP with constraints: [A] / [CO3]

$7x + 8y \geq 168$

$2x + 4y \geq 60$

$x, y \geq 0$

15. Find the marginal and average cost if the total cost is [A] / [CO4]

$TC = 0.5Q - Q^2.$

16. Maximize the total profit $\pi = -2Q^3 - 15Q^2 + 3000Q - 1200.$ [A] / [CO4]

[6x5 = 30]

Part C

{Essay Questions}

Answer any **3** questions

Each question carries **10** marks

17. A car depreciates by 40% in the first year, 30% in the second year and 20% thereafter. I buy a car for $14,700 when it is 2 years old. [E] / [CO1]

(a) How much did it cost when new?

(b) After how many years will it be worth less than 25% of the amount that I paid for it?

18. Solve the system of equations a) using elimination method b) using substitution method [U] / [CO2]

$$4x - 3y = 22$$
$$7x - 6y = 34$$

19. Find a) the level of income and the rate of interest that concurrently bring equilibrium to the economy and b) estimate the level of consumption C , investment I, the transaction-precautionary demand for money M_t , and the speculative demand for money M_w when the money supply M_s = 275 and [U] / [CO2]

$$C = 89 + 0.6Y \qquad I = 120 - 150i$$
$$M_t = 0.1Y, \qquad M_w = 240 - 250i,$$

20. Solve the LPP [A] / [CO3]

Maximize $z = 50x_1 + 30x_2$

subject to

$$2x_1 + x_2 \leq 14$$
$$5x_1 + 5x_2 \leq 40$$
$$x_1 + 3x_2 \leq 18$$
$$x_1, x_2 \geq 0$$

21. Optimize the function $y = -3(x-4)^6$. [An] / [CO4]

22. Optimize the function $y = (6-x)^3$. [An] / [CO4]

[10x3 = 30]